CPSC 457 - Assignment 2

Scheduling Algorithms Reflection

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Each scheduler was compiled and installed in the supplied Minix 3.1.0 disk image, booted in QEMU, and evaluated with ten runs of the Assignment 1 program using the arguments 5 30 10 40 5 35 20. The averages use the real elapsed time reported by the time command. FCFS averaged **0.007 seconds**. Its short runtime is consistent with the expectation that retaining a running process until it blocks avoids round-robin rotation, although these measurements do not establish an overhead advantage. Round robin averaged **0.005 seconds**. It had the smallest measured average, but its main expected benefit is sharing CPU time fairly among runnable processes at the same priority, rather than necessarily reducing total completion time. MLFQ averaged **0.008 seconds**. It demotes processes that consume a full quantum and promotes processes that block early toward their original priority; its expected responsiveness benefit is more relevant to mixed workloads with longer CPU bursts. The small differences here are consistent with all three policies handling this brief workload quickly. The Assignment 1 implementation calculates Fibonacci numbers iteratively, so the requested inputs finish very quickly. Individual real-time measurements were displayed in increments of 0.01 seconds, including values of 0.00 seconds. Averaging those readings does not create additional measurement precision, and these results do not support a reliable ranking of the schedulers. All thirty runs completed successfully and produced the expected Fibonacci values. An operating system can function with FCFS when processes regularly block or finish, but pure non-preemptive FCFS is a poor general-purpose choice: a CPU-bound process can delay other ready processes indefinitely, making interactive programs and services unresponsive. In this implementation, Minix retains its separate priority queues and default scheduling for kernel tasks, which makes it different from applying one global FCFS queue to the entire operating system.